

JK LakshmiPat University Jaipur

INSTITUTE OF ENGINEERING AND TECHNOLOGY

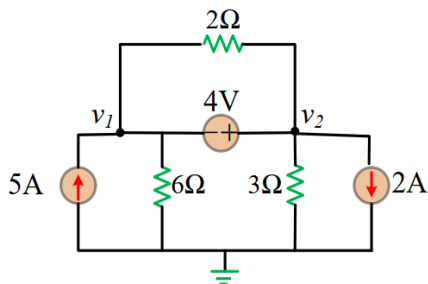
Practice sheet 1 B. Tech. All, Semester I

Course EE1118 Fundamental of Electrical and Electronic Engineering

1. Calculate Node Voltages in the given circuit.

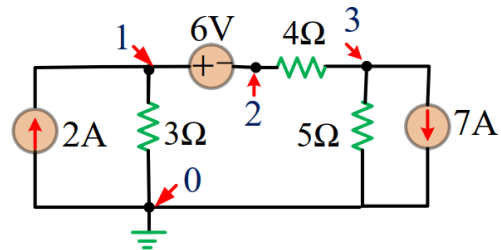
$$v_1 = 13.33\text{V and } v_2 = 20\text{V}$$

2. Calculate Node Voltages in the given circuit.



$$v_1 = 3.33\text{ V and } v_2 = 7.33\text{ V}$$

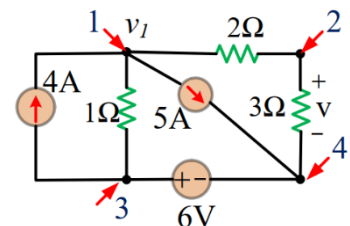
3. Calculate current through the 4Ω resistor in the given circuit.



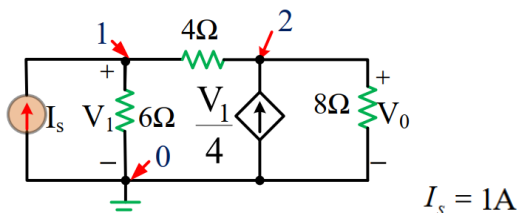
$$v_1 = -2.77\text{V}, v_2 = -8.77\text{V and } v_3 = -20.43\text{V } I = 2.91\text{ A}$$

4. Solve for voltage V across the 3Ω resistor using node voltage analysis for the given circuit.

$$v_1 = 4.1667\text{V } \text{voltage across } 3\Omega \text{ resistor is } = 2.5\text{V}$$

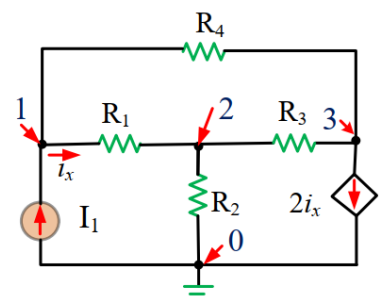


5. Find I_s for the circuit shown. Take $V_0 = 16\text{V}$.



$$I_s = 1\text{A}$$

6. Determine the node voltages and the current i_x for the given circuit having 4 nodes with the following values, $R_1 = 2\Omega$, $R_2 = 1\Omega$, $R_3 = 5\Omega$, $R_4 = 6\Omega$ and $I_1 = 6\text{A}$.



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Practice sheet 2

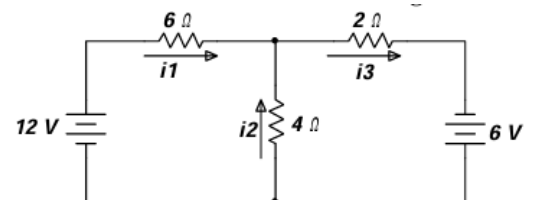
B. Tech. All, Semester I

Course EE1118 Fundamental of Electrical and Electronic Engineering

Mesh Analysis

1. Determine the currents in the following circuits with reference to the indicated direction.

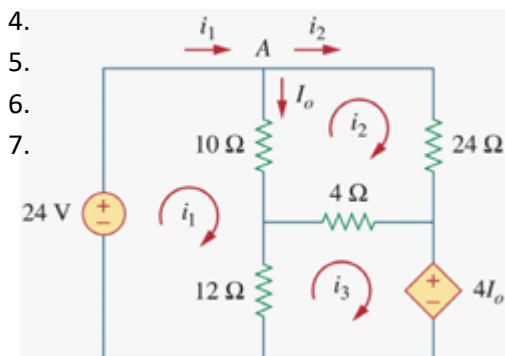
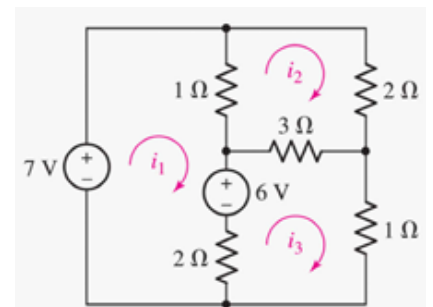
$$i_1 = 2.18 \text{ A}, \quad i_2 = 0.27 \text{ A} \quad \text{and} \quad i_3 = 2.45 \text{ A}$$



2. For the show circuit, find the mesh currents using mesh analysis.

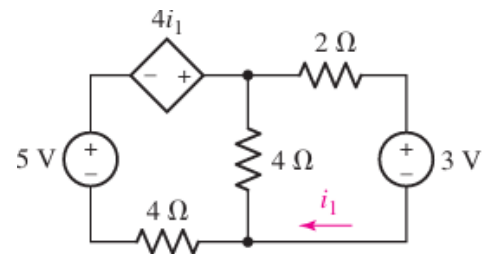
$$i_1 = 3 \text{ A}, \quad i_2 = 2 \text{ A} \quad \text{amd} \quad i_3 = 3 \text{ A}.$$

3. Use mesh analysis to find the current in the circuit of the shown figure.

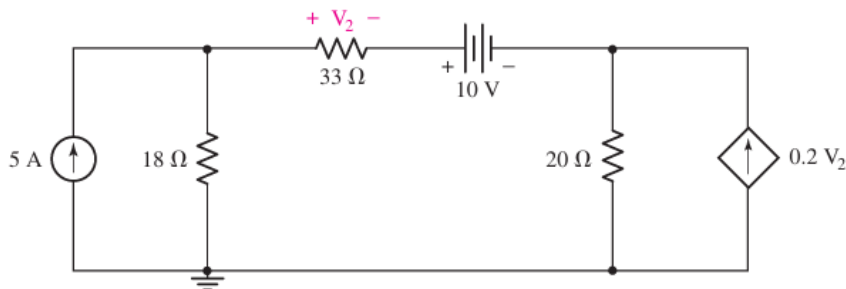


$$i_1 = 2.25 \text{ A}, \quad i_2 = 0.75 \text{ A} \quad \text{and} \quad i_0 = 1.5 \text{ A}.$$

4. Determine the current i_1 in the given circuit.



5. Determine the voltage v_2 of the given circuit.



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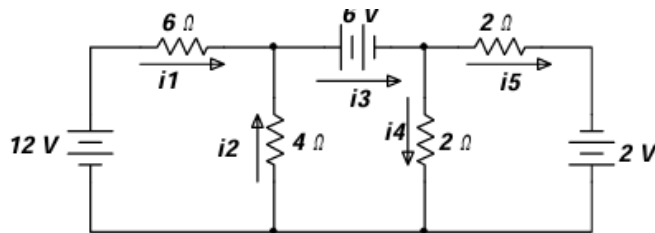
Practice sheet 3

B. Tech. All, Semester I

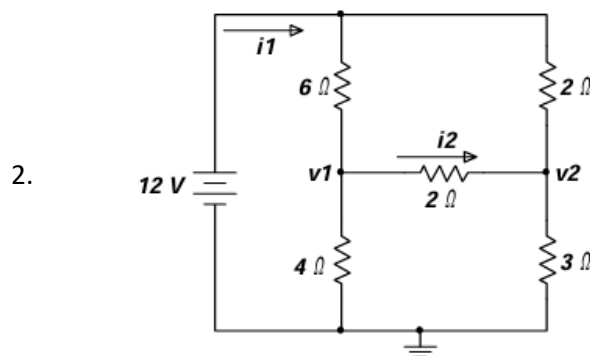
Course EE1118 Fundamental of Electrical and Electronic Engineering

Super Node and Super Mesh Analysis

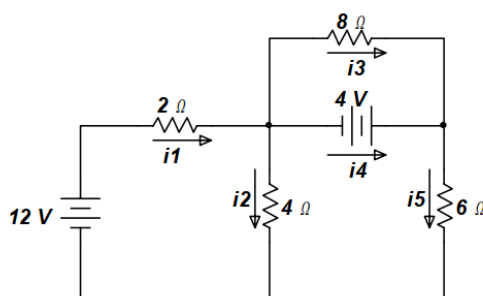
Determine the currents in the following circuits with reference to the indicated direction.



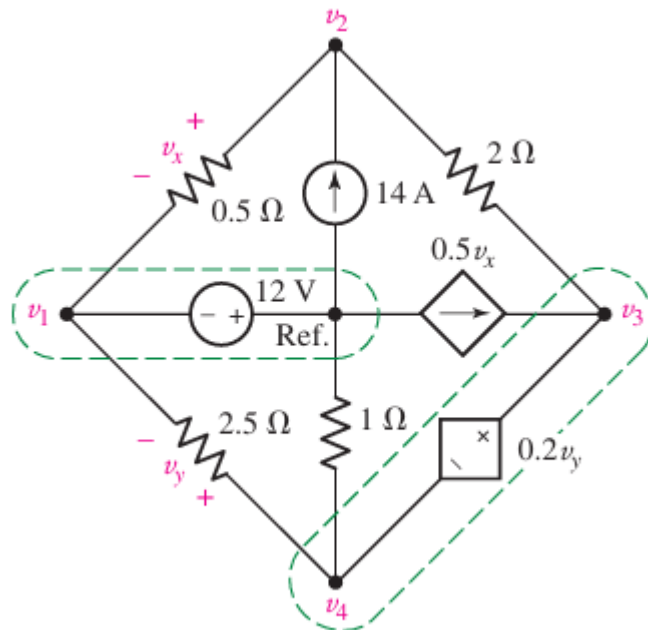
1. Answer: $i1 = 1.180A$, $i2 = -1.240A$, $i3 = -0.058A$, $i4 = -0.529A$, $i5 = 0.471A$



2. Answer: $i1 = 3.690A$, $i2 = -0.429A$, $v1 = 5.83V$, $v2 = 6.69V$



3. Answer: $i1 = 3.09A$, $i2 = 1.45A$, $i3 = -0.50A$, $i4 = 2.14A$, $i5 = 1.64A$



3.

$v_1 = -12 \text{ V}$, $v_2 = -4 \text{ V}$, $v_3 = 0 \text{ V}$, and $v_4 = -2 \text{ V}$.

4. Determine the voltage labelled v in the circuit.

